Climate Technology and Mitigation Deterrence

How Direct Air Capture Narratives Shape Public Expectations

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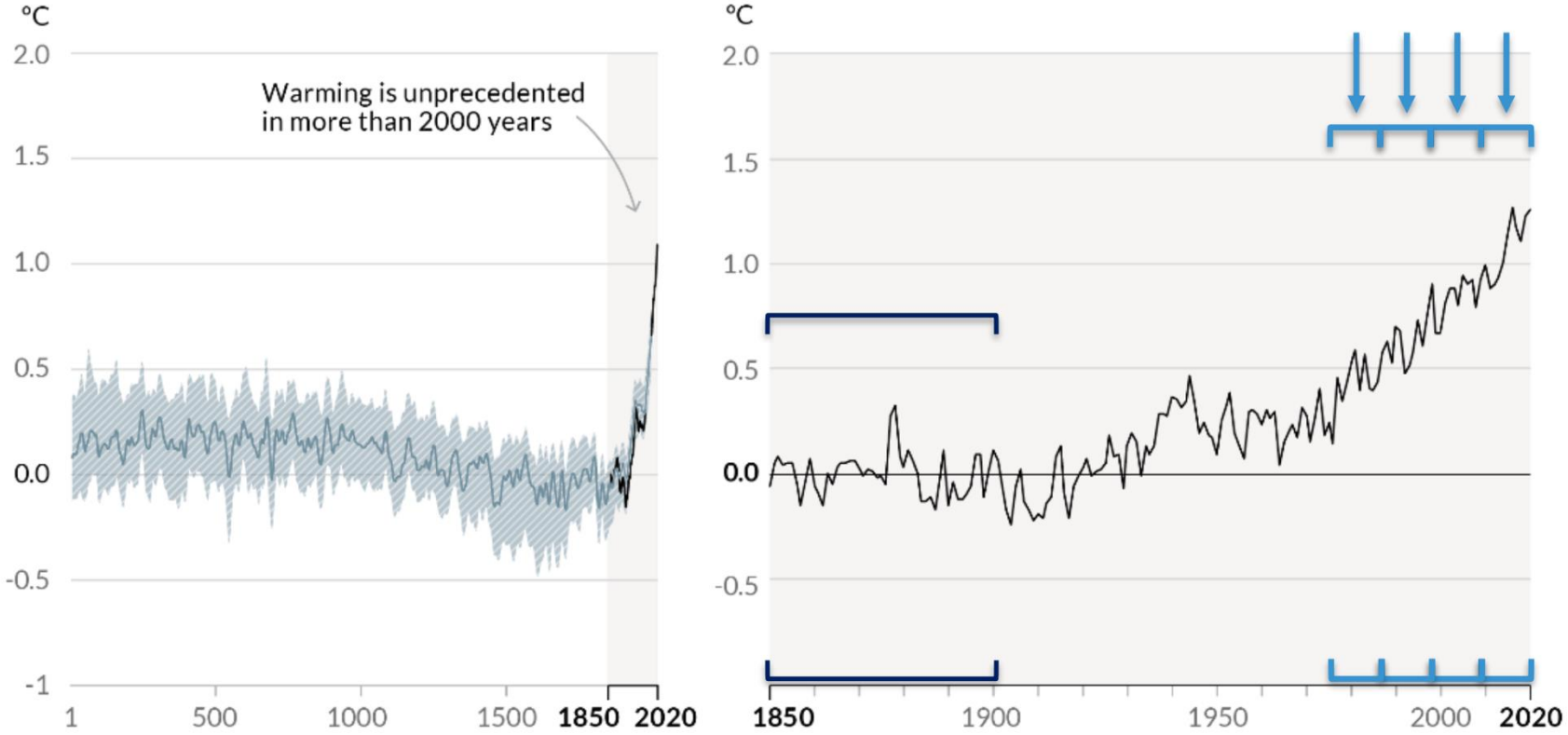
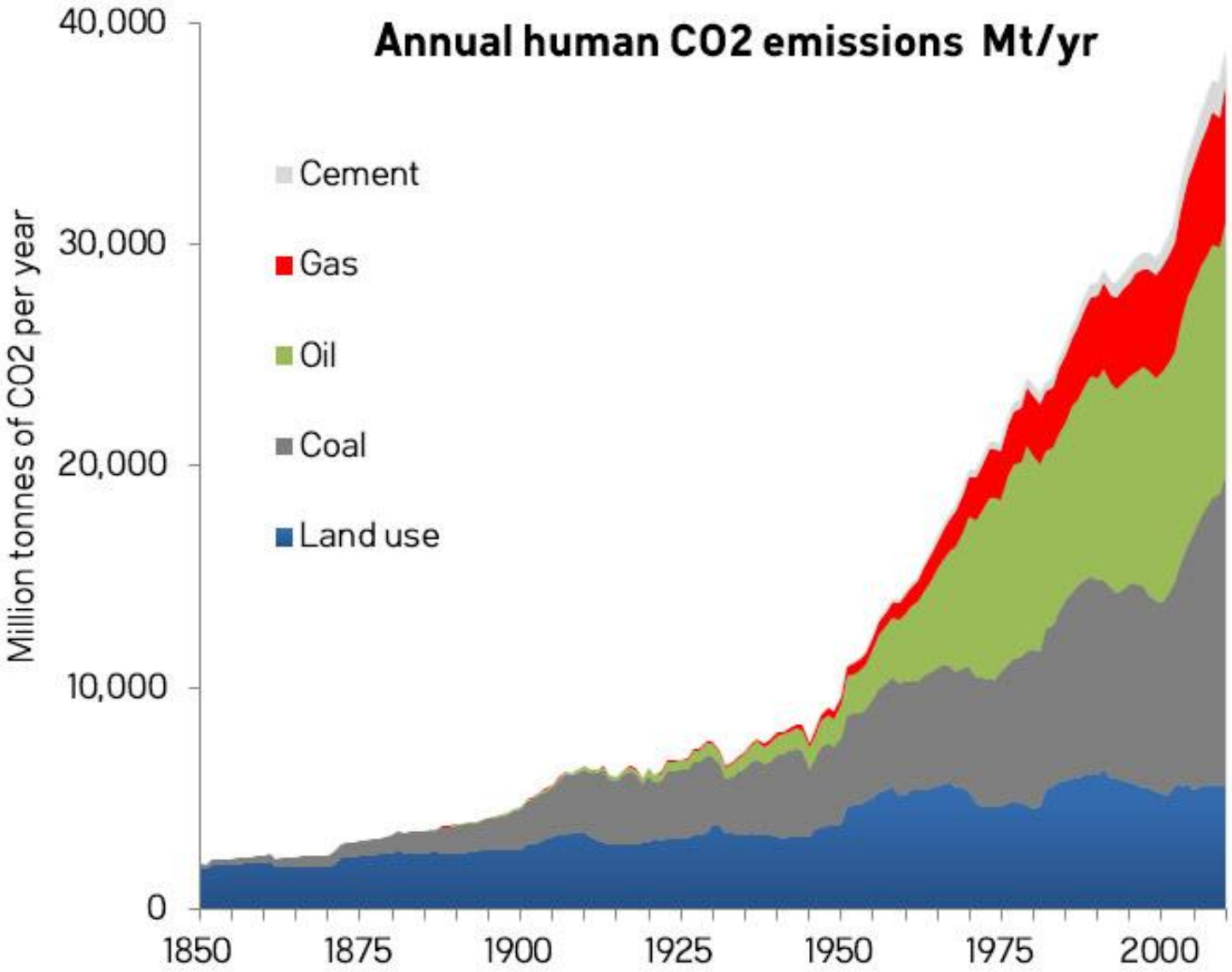
Jan 26, 2026

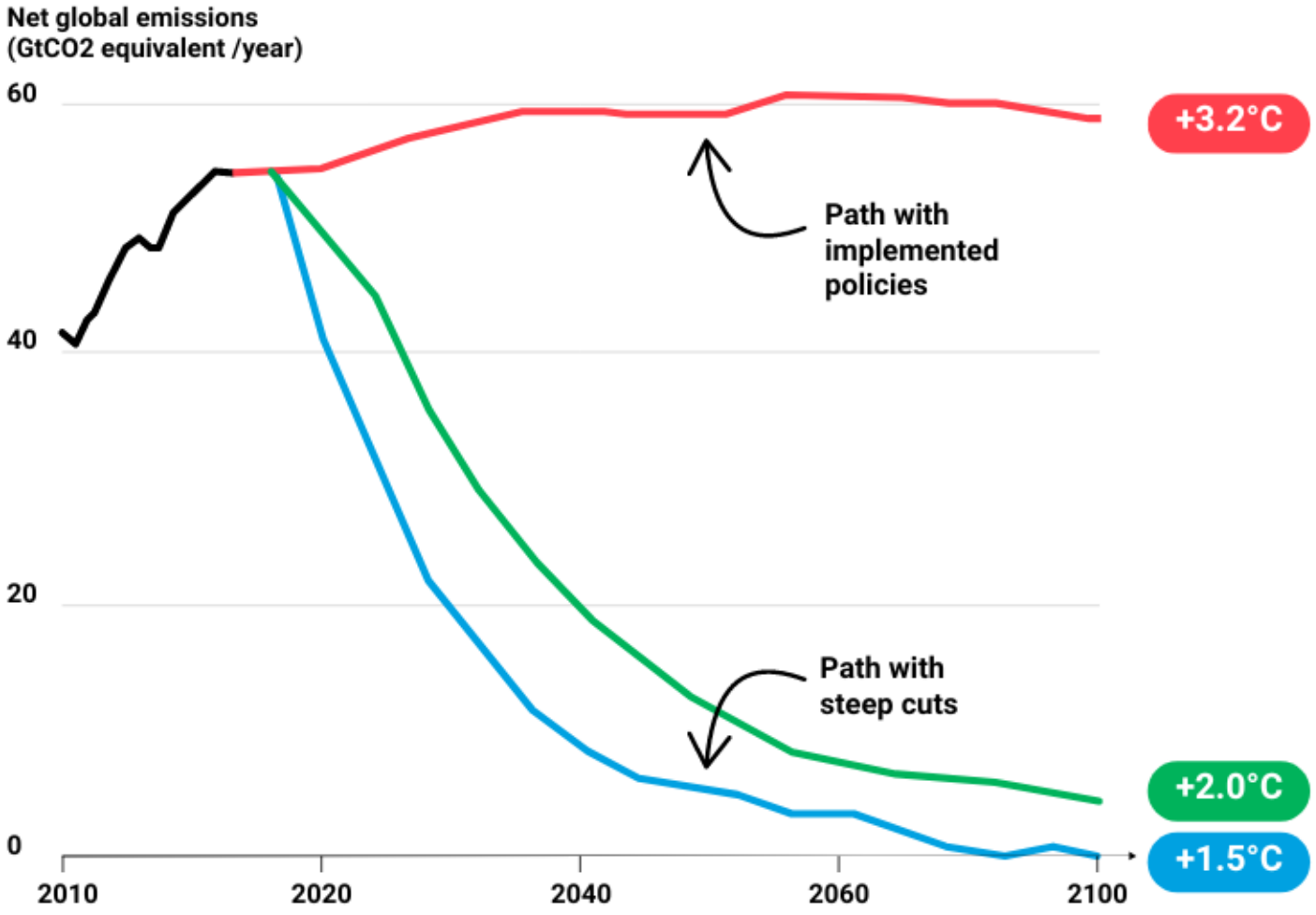
Human influence has warmed the climate at a rate that is unprecedented in at least the last 2000 years

Changes in global surface temperature relative to 1850-1900




Annual human CO2 emissions Mt/yr
Cement
Gas
Oil
Coal
Land use
Million tonnes of CO2 per year
40,000
30,000
20,000
10,000
0
1850
1875
1900
1925
1950
1975
2000

If we act now, steep cuts to greenhouse gas emissions can limit global warming



Source: IPCC, 2023: Summary for Policymarkers. In: Climage Change 2023: Synthesis Report. A Report of the Intergovernmental Panel on Climate Change.

Climate Actions

- DAC
- Afforestation
- Soil carbon sequestration
- ……

Carbon Dioxide Removal (CDR)

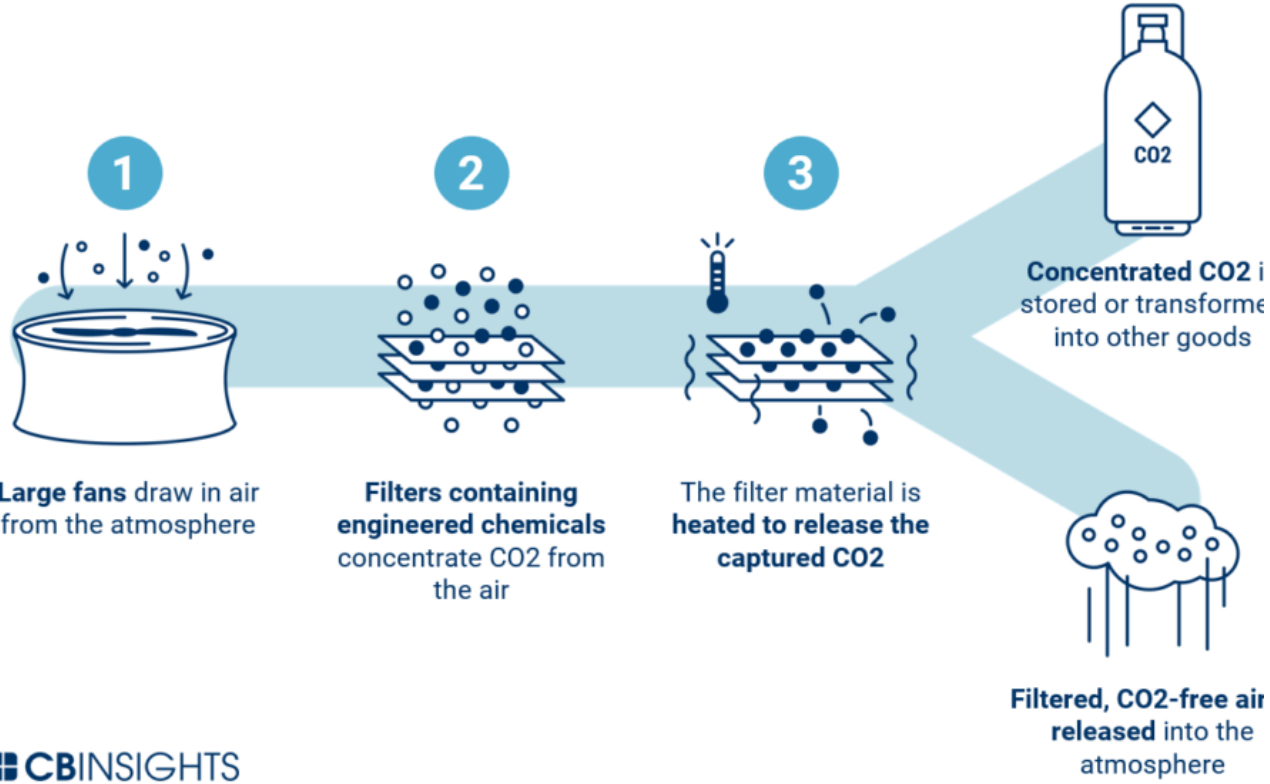
- Renewables
- Energy efficiency
- Lifestyle change
- ……

Conventional Mitigation

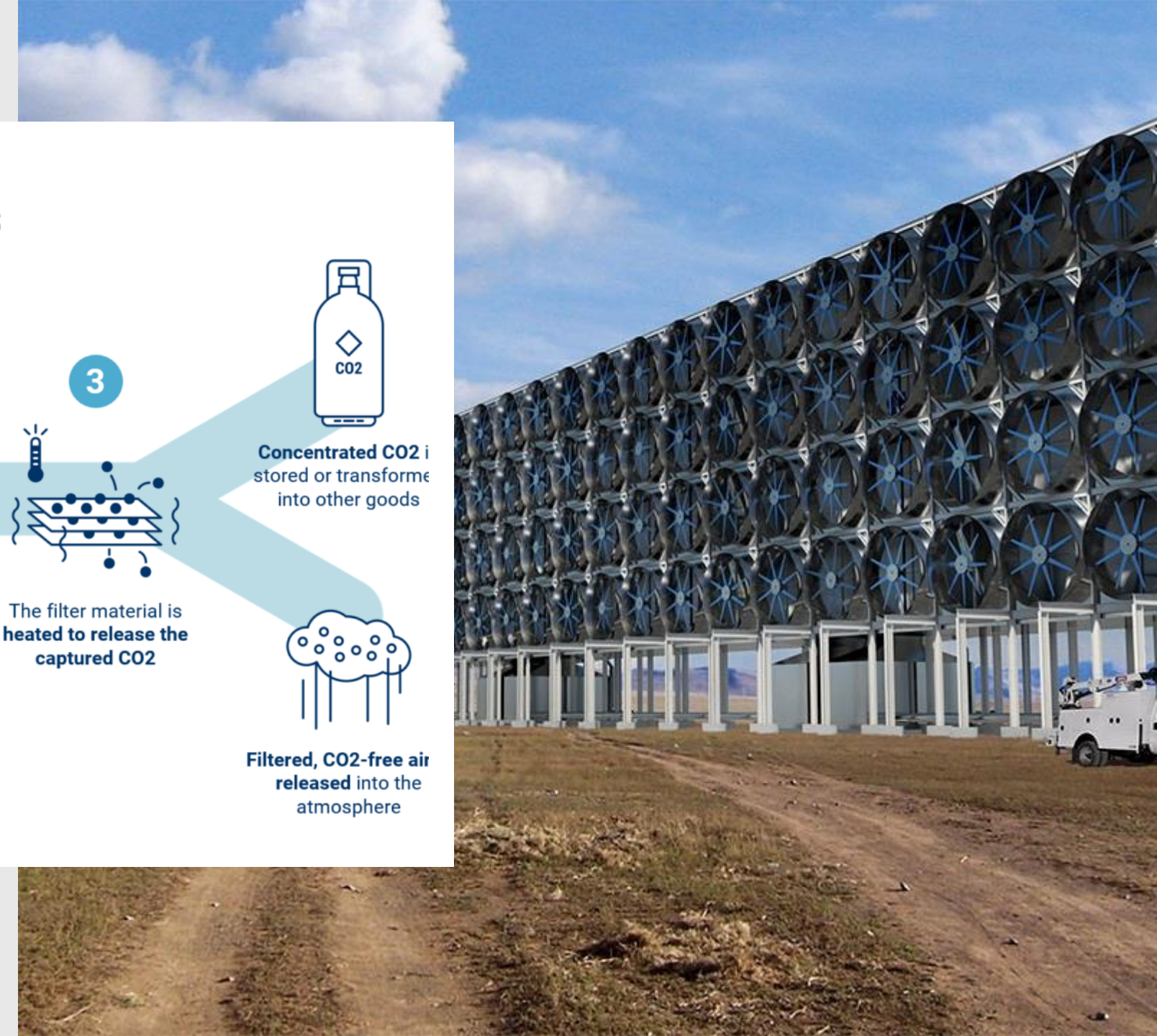
- Climate-resilient infrastructure
- Agriculture
- Preparedness
- ……

Adaptation

How direct air capture works



CBINSIGHTS

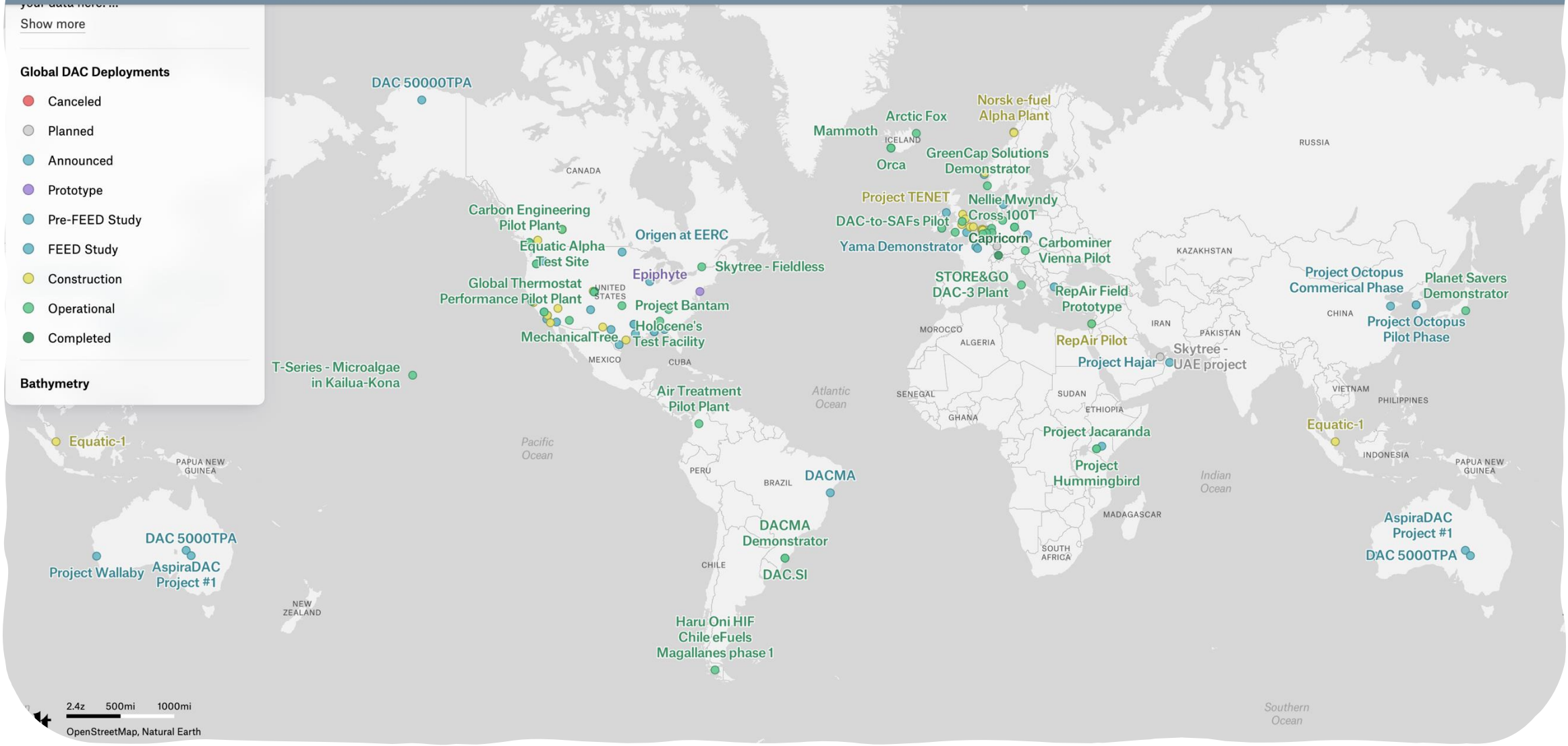


Show more

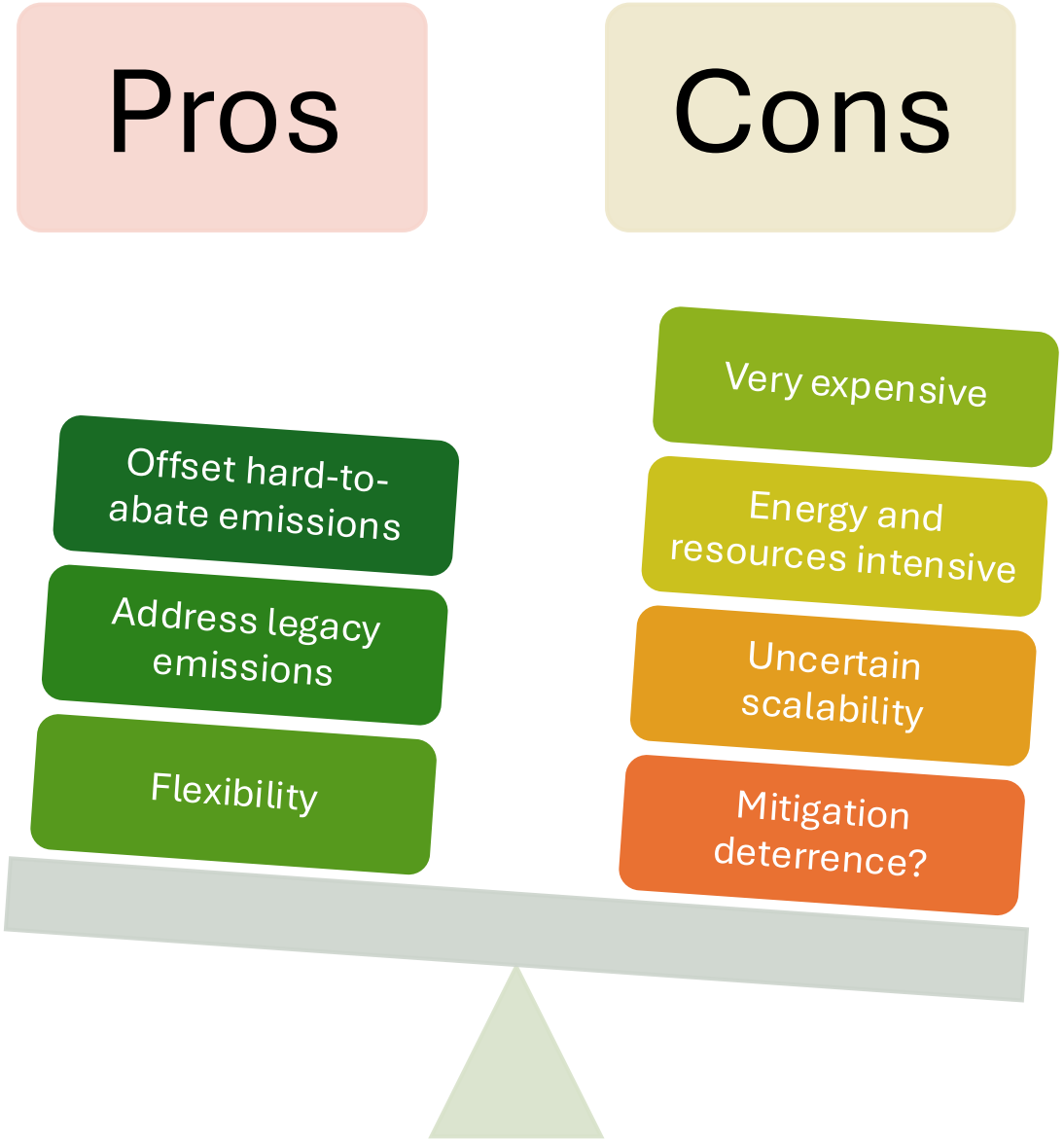
Global DAC Deployments

- Canceled
- Planned
- Announced
- Prototype
- Pre-FEED Study
- FEED Study
- Construction
- Operational
- Completed

Bathymetry



Debate over DAC



Mitigation Deterrence	refers to the concern that emphasizing DAC may reduce urgency or support for cutting emissions,
	by shifting attention, political capital, or resources away from conventional mitigation.

Concern Not Unfounded

US

- USD 3.5 billion to fund two projects in Texas & Louisiana
- 45Q tax credit raised to USD 180/t CO_2 for DAC (Inflation Reduction Act (2022))
- ……

UK

- GBP 20 billion (≈ USD 25 bn) for carbon capture, including direct air capture
- ……

Unsettled Debate Over Mitigation Deterrence

EMPIRICAL EVIDENCE IS MIXED

CONFUSING MEANS WITH GOALS

SOME SUBSTITUTION IS OPTIMAL

Two Analogies

Seat Belt

Adaptation

What Actually Matters

Degree of Deterrence

- Not all deterrence is the same

Narratives

- Optimistic
- Balanced/Cautious

This project asks two questions:

How—and to what extent—do different DAC narratives change people's perceived urgency of conventional mitigation?

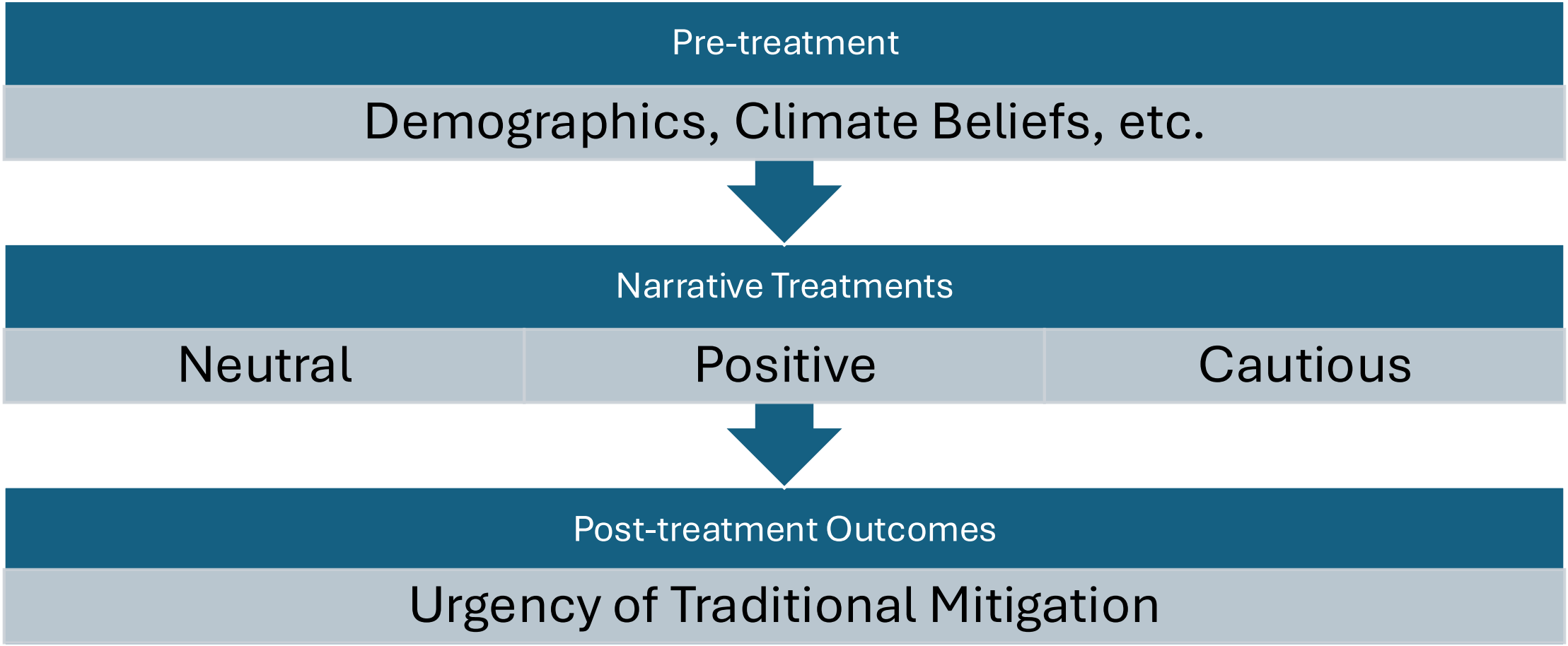
Under different DAC narratives, how do people evaluate fossil-fuel firms' DAC investments, and how does this shape support for regulatory intervention?

Sample

About 3,000 respondents stratified by gender, age, and tiers of cities

Administered by Dynata

Survey Process



Narrative Treatments

Neutral

Direct Air Capture (DAC) is a technology that uses chemical or physical methods to directly capture carbon dioxide from the air. The captured carbon dioxide can be stored underground or used for industrial purposes. At present, DAC is still at an early stage of development, with only a small number of demonstration projects operating worldwide.

Positive

Neutral +

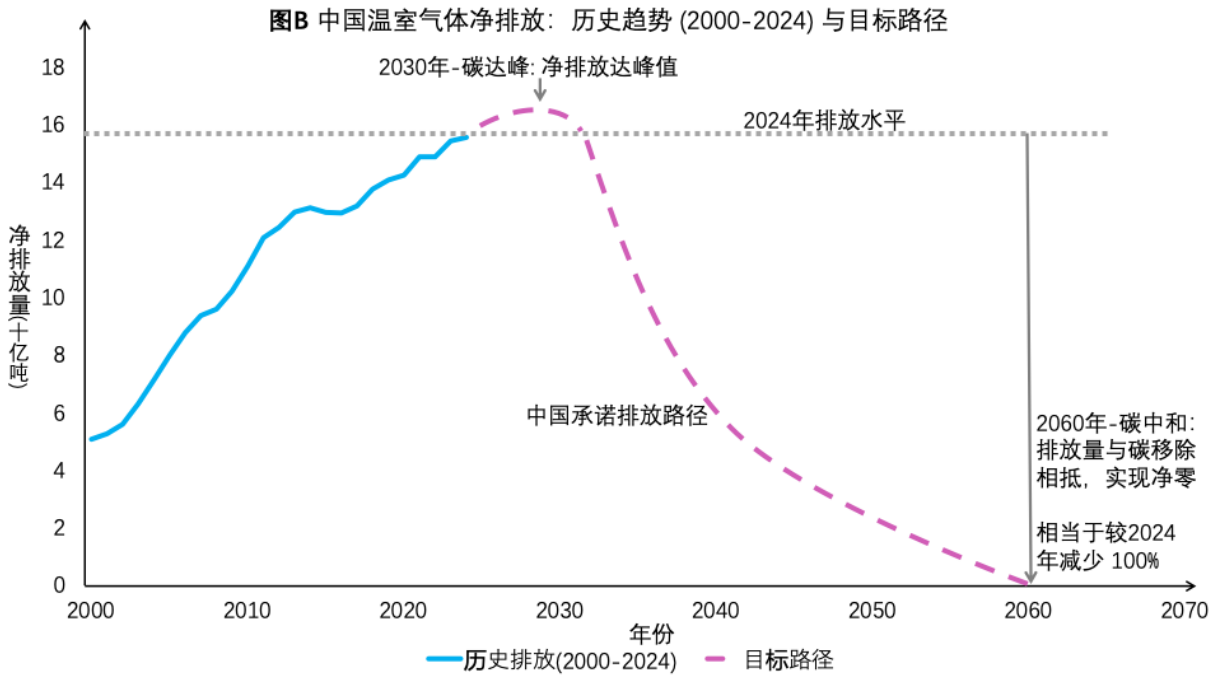
Some argue that DAC is a highly promising technology. It can capture carbon dioxide from the air without being constrained by the location of emissions. As technology advances and costs decline, DAC is expected to be deployed at scale in the future, offering a new option for addressing climate change.

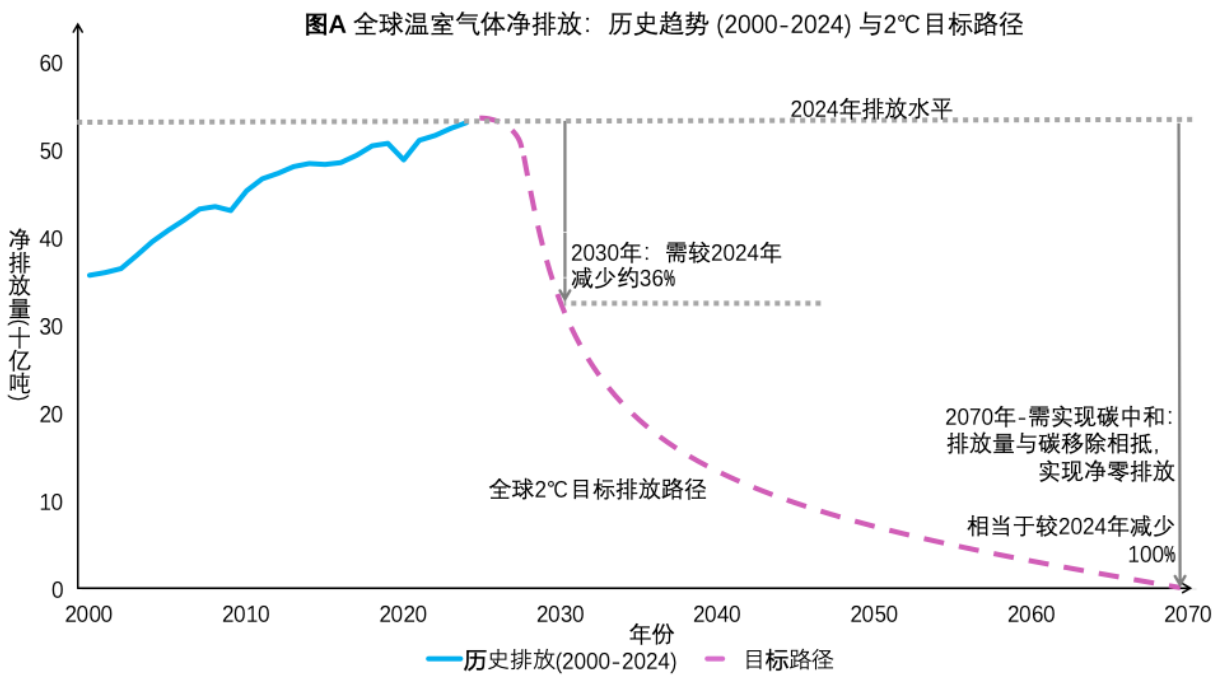
Cautious

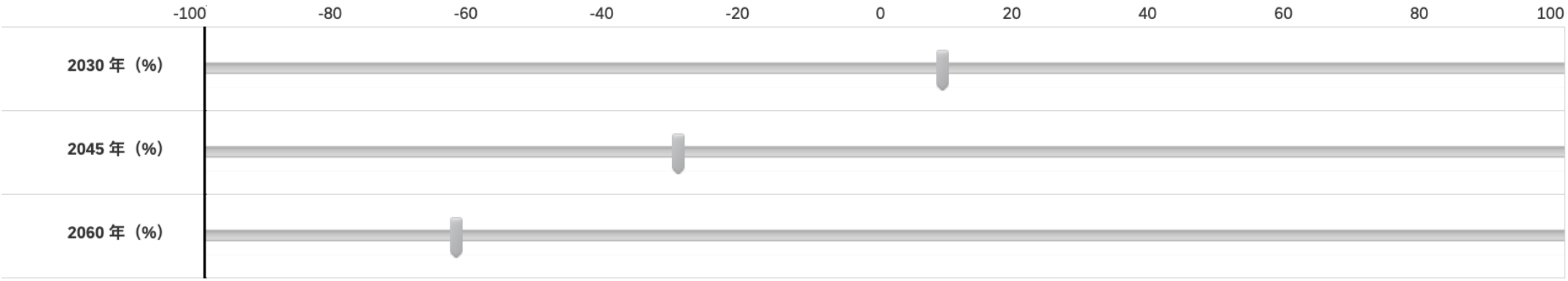
Neutral +

Some argue that DAC may play a role in offsetting emissions from hard-to-abate sectors such as aviation and heavy industry. However, DAC currently remains costly and energy-intensive, and large-scale deployment faces significant challenges. Therefore, DAC is better viewed as a complementary measure for addressing climate change rather than a substitute for reducing emissions at the source.

Measuring Mitigation Urgency: Background Information

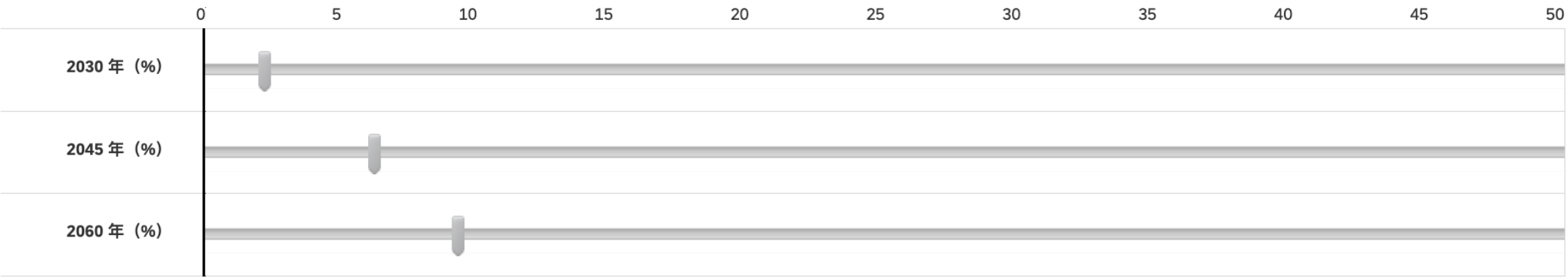






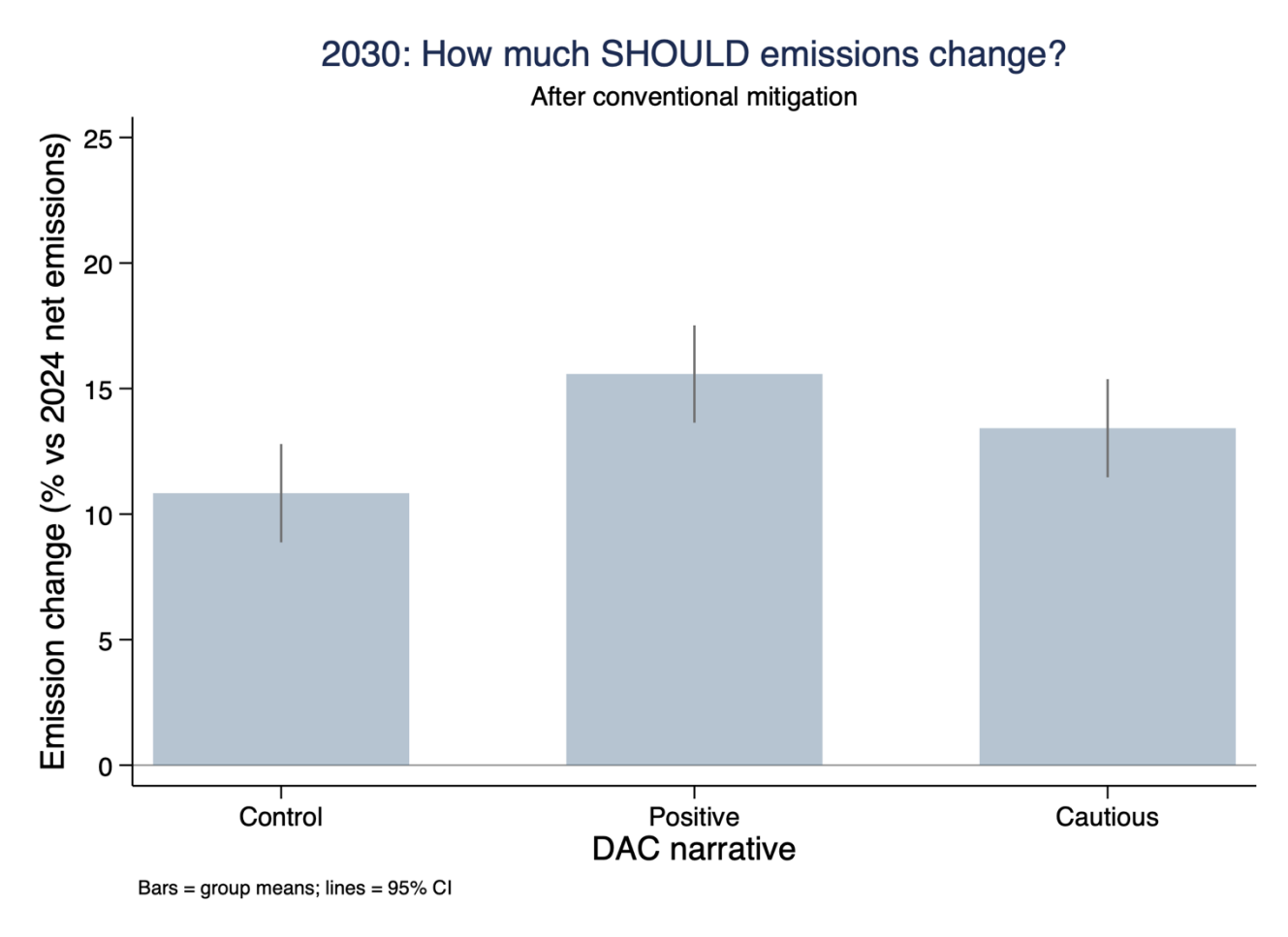
Measuring Mitigation Urgency

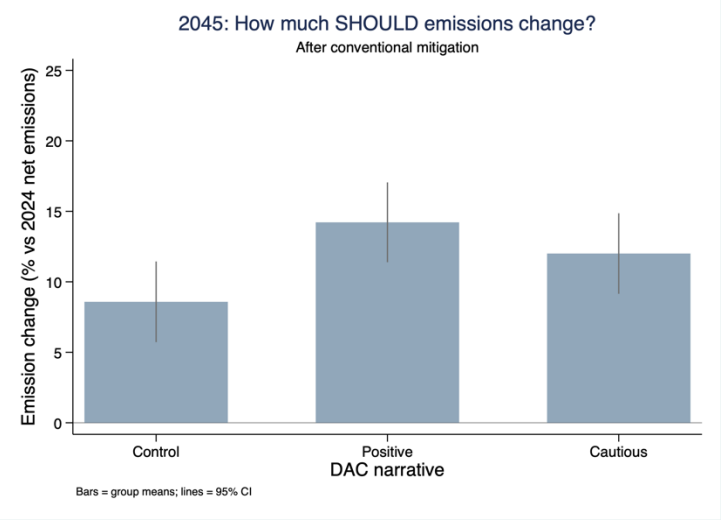
A. The emission changes China should achieve after conventional mitigation *(relative to the 2024 net emissions level)*



Measuring Carbon Removal

B. Through carbon removal, the additional reduction in emissions that China should achieve


2030: How much SHOULD emissions change?
After conventional mitigation
Emission change (% vs 2024 net emissions)
25
20
15
10
5
0
Control
Positive
Cautious
DAC narrative
Bars = group means; lines = 95% CI


2045: How much SHOULD emissions change?
After conventional mitigation
Emission change (% vs 2024 net emissions)
25
20
15
10
5
0
Control
Positive
Cautious
DAC narrative
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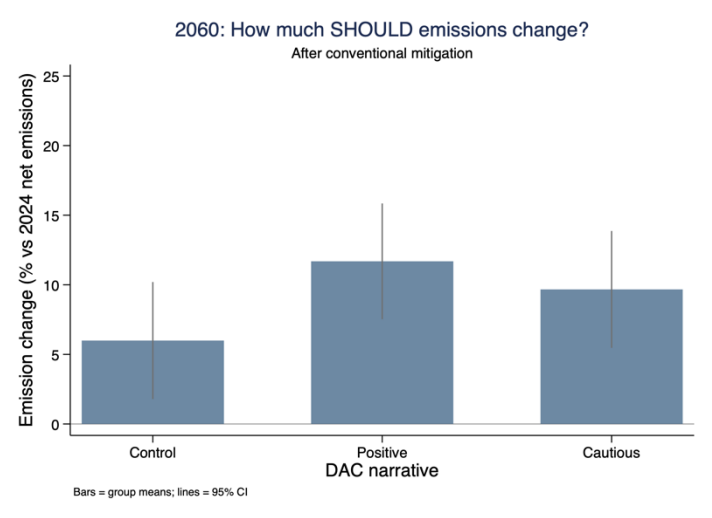
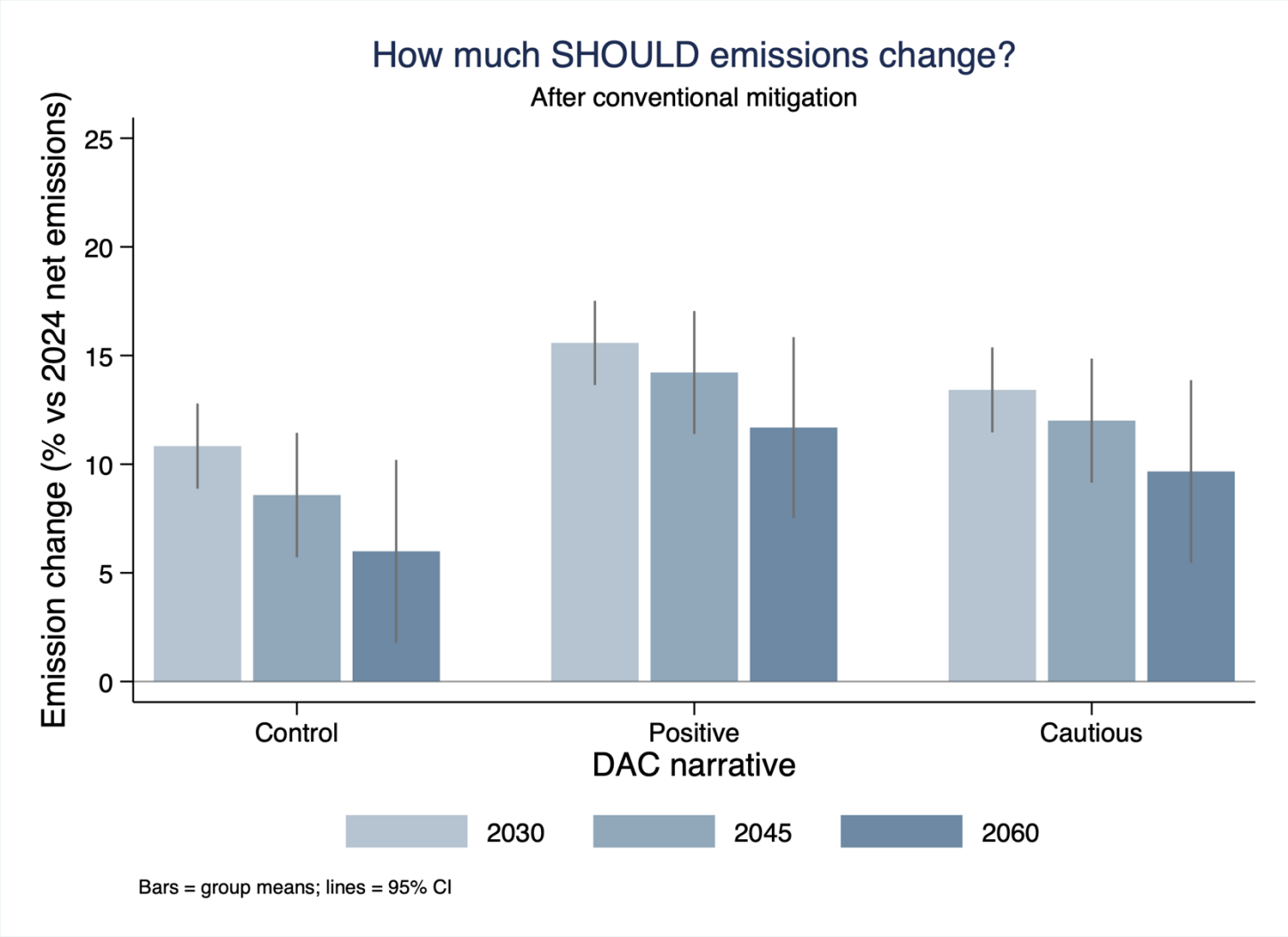
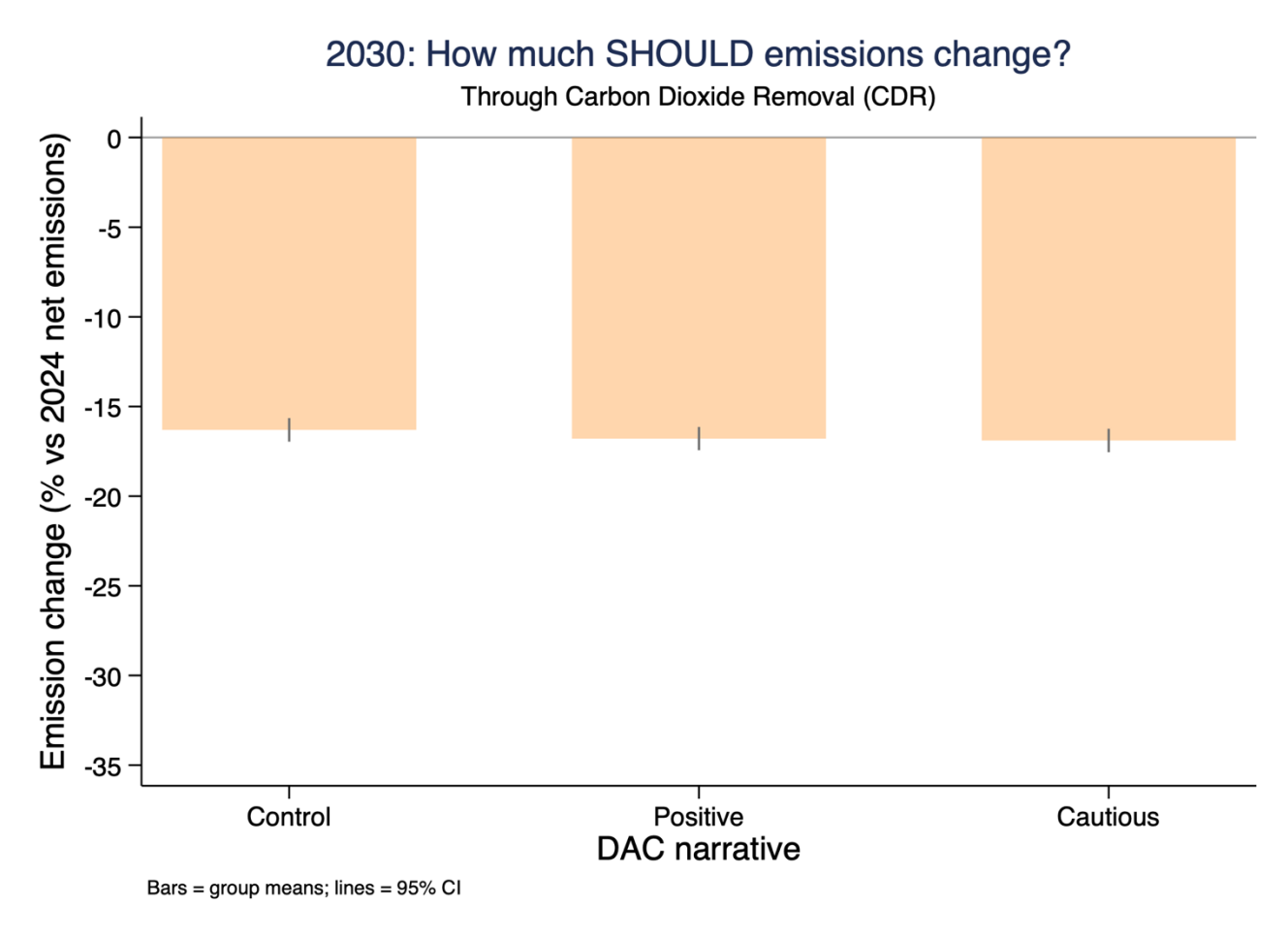
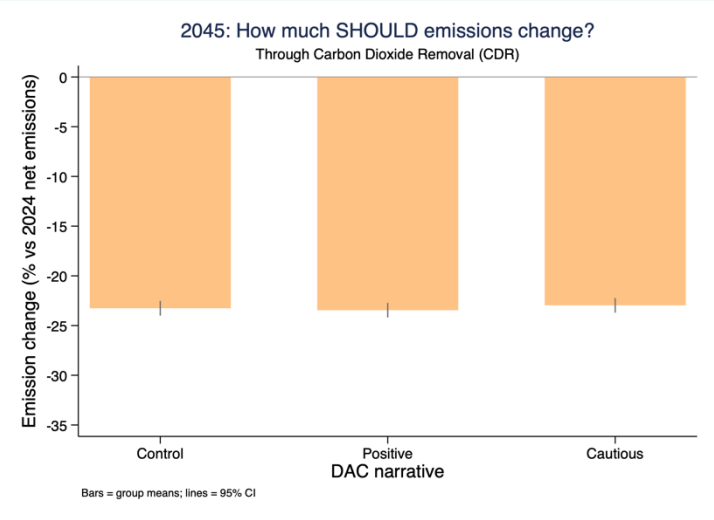

2060: How much SHOULD emissions change?
After conventional mitigation
Emission change (% vs 2024 net emissions)
25
20
15
10
5
0
Control
Positive
Cautious
DAC narrative
Bars = group means; lines = 95% CI

- *DAC narratives generate modest mitigation deterrence.*




2030: How much SHOULD emissions change?
Through Carbon Dioxide Removal (CDR)
Emission change (% vs 2024 net emissions)
0
-5
-10
-15
-20
-25
-30
-35
Control
Positive
Cautious
DAC narrative
Bars = group means; lines = 95% CI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Emission change (% vs 2024 net emissions)
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-10
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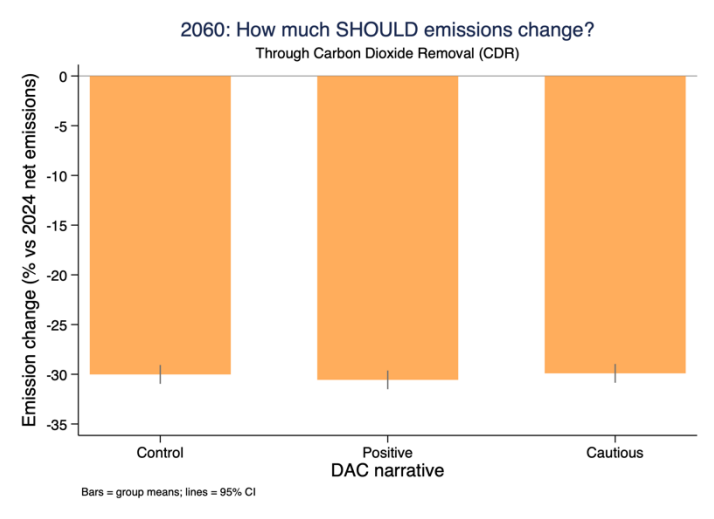
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Through Carbon Dioxide Removal (CDR)
Emission change (% vs 2024 net emissions)
0
-5
-10
-15
-20
-25
-30
-35
Control
Positive
Cautious
DAC narrative
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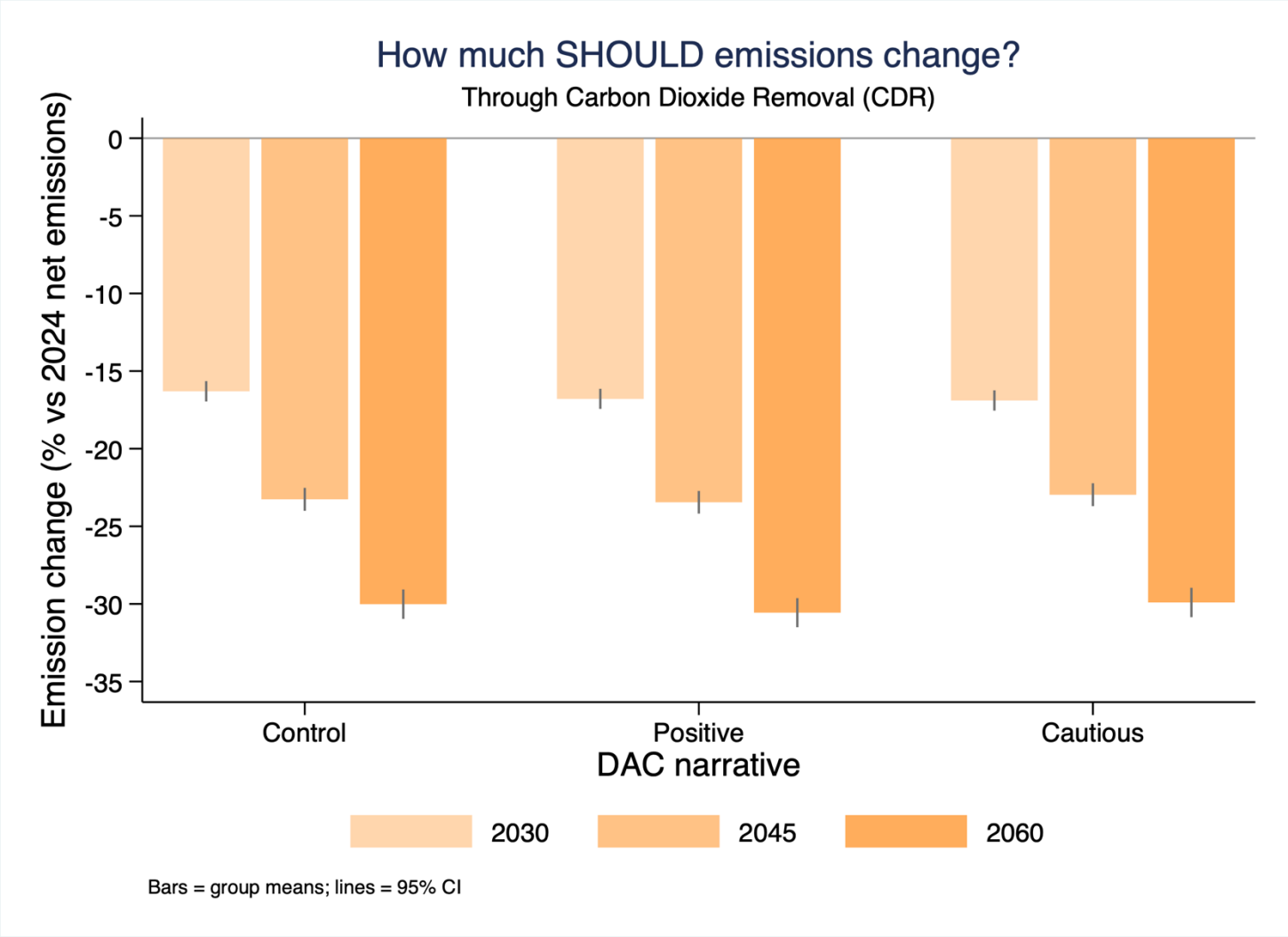
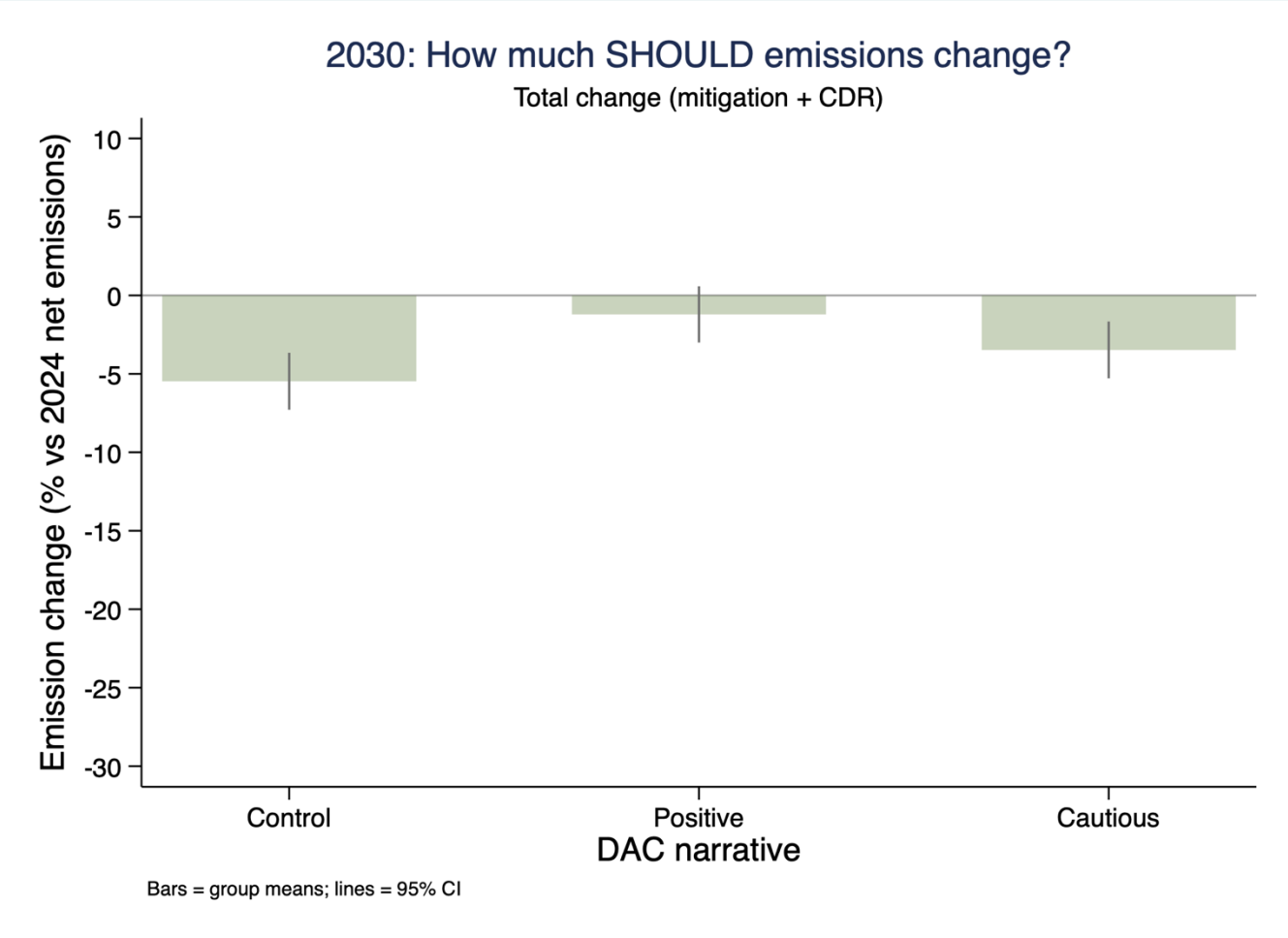
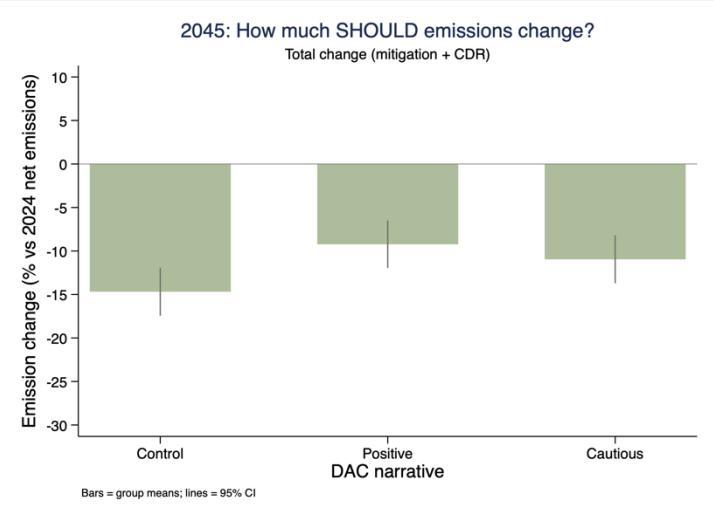
- *DAC narratives do not affect beliefs about how much emissions reduction should be achieved through carbon dioxide removal.*




2030: How much SHOULD emissions change?
Total change (mitigation + CDR)
Emission change (% vs 2024 net emissions)
10
5
0
-5
-10
-15
-20
-25
-30
Control
Positive
Cautious
DAC narrative
Bars = group means; lines = 95% CI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-15
-20
-25
-30
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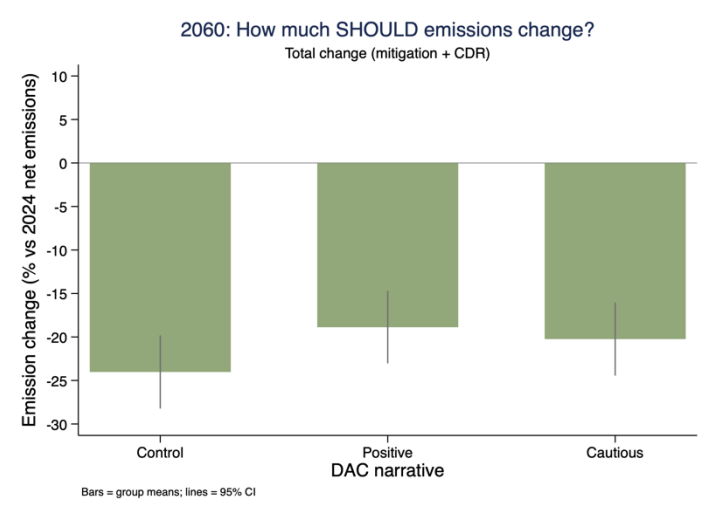
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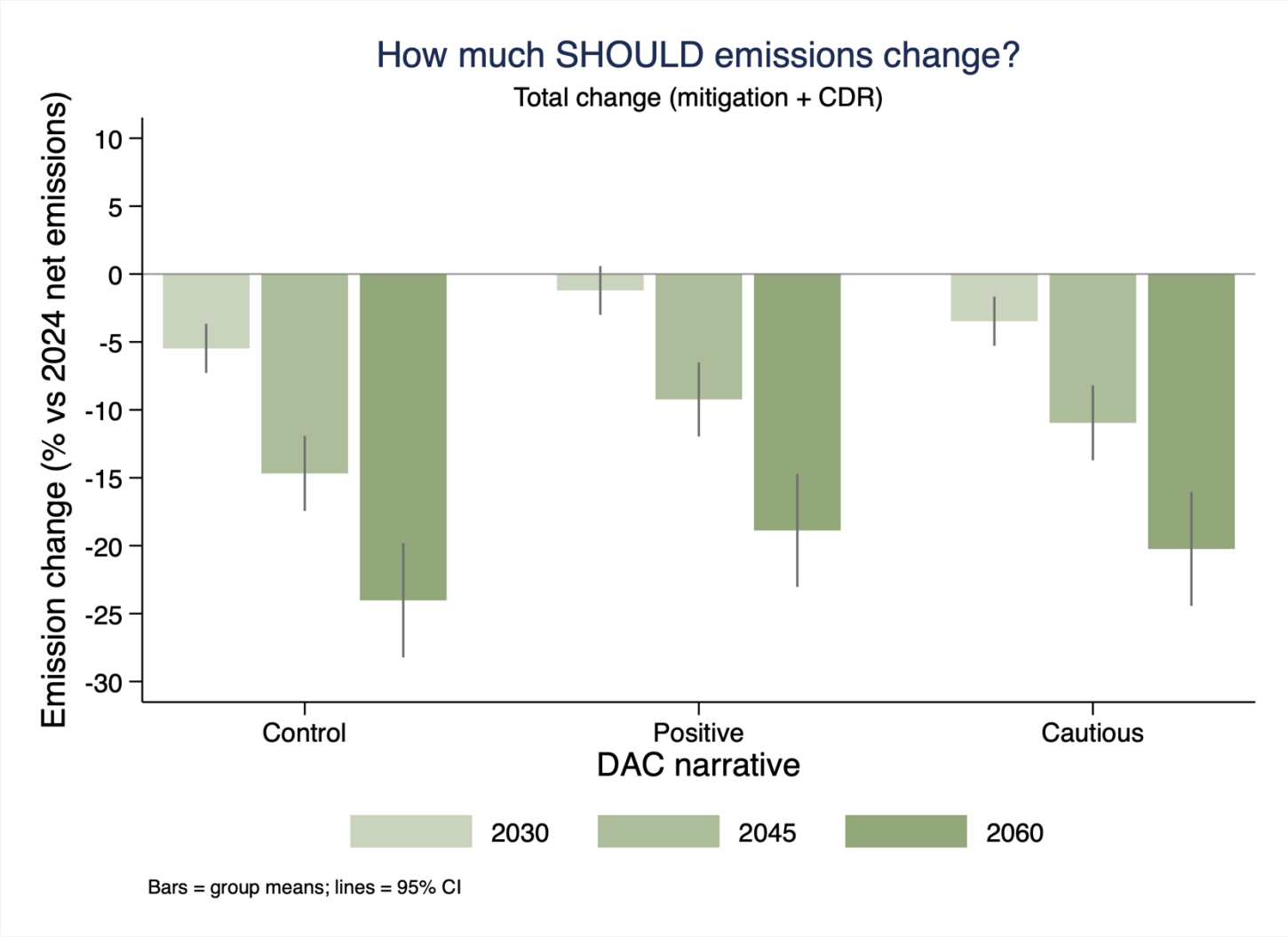
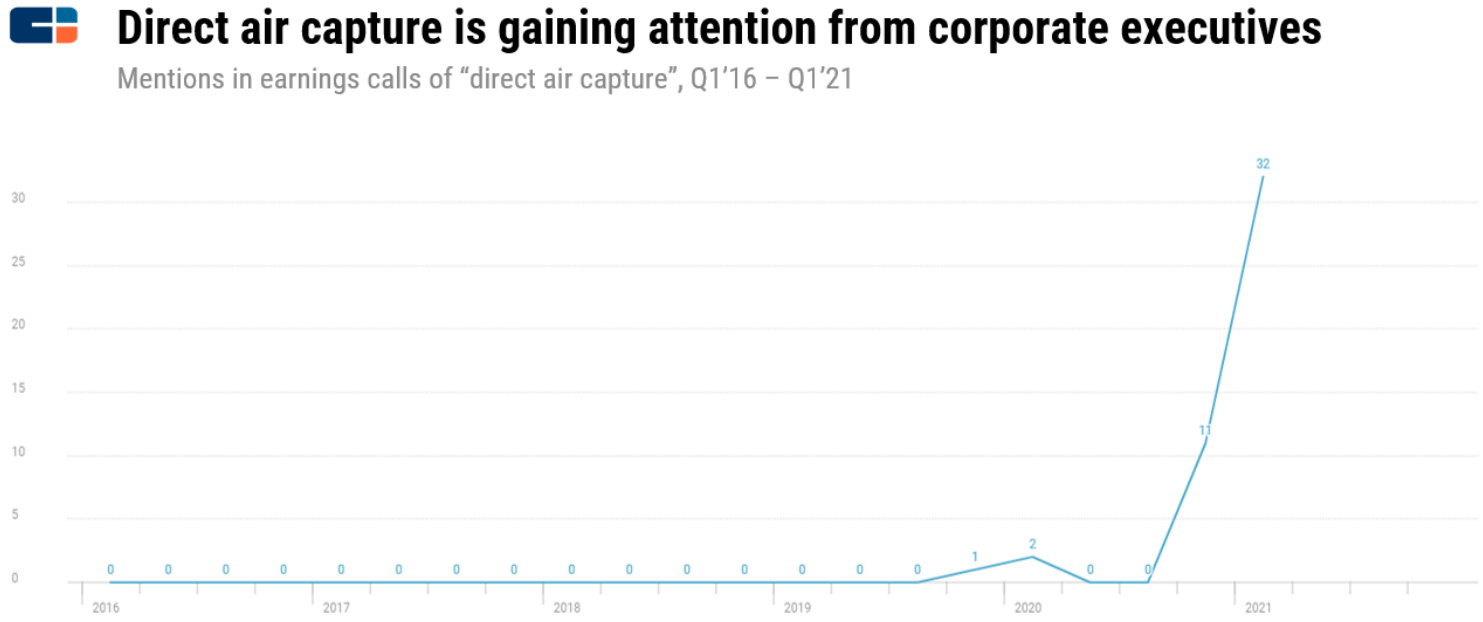
0
-5
-10
-15
-20
-25
-30
Control
Positive
Cautious
DAC narrative
Bars = group means; lines = 95% CI

- *Across all groups, respondents' preferred emissions reductions fall well short of China's official targets and IPCC-recommended pathways.*
- *Overall ambition is substantially lower than what current climate goals require.*



DAC as Greenwashing?

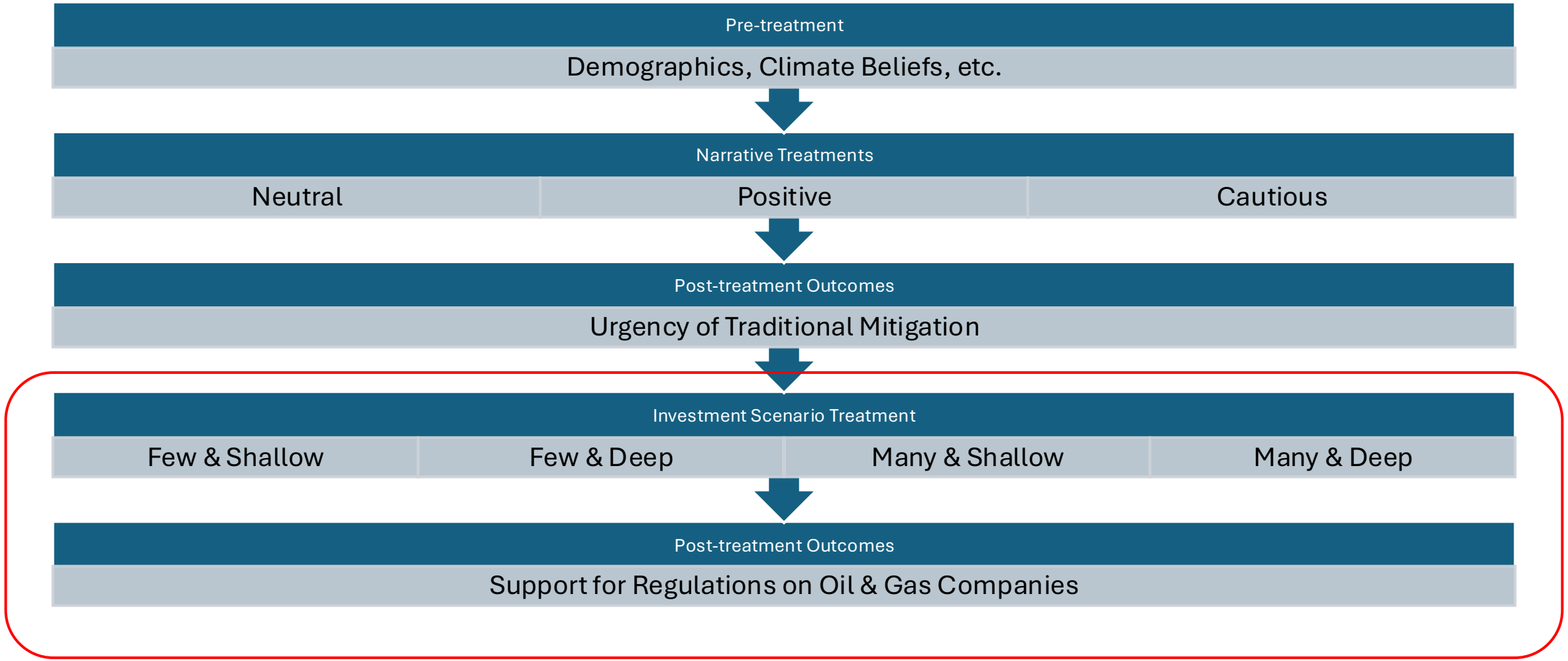


Source: cbinsights.com .

Second Research Question

- *Do oil and gas companies' DAC investments increase or reduce public support for strict regulation—and does this depend on the DAC narrative?*

Survey Process



Investment Scenarios

Few vs. Many

A small proportion (≈15%) of oil and gas companies invest in carbon removal technologies such as Direct Air Capture (DAC).

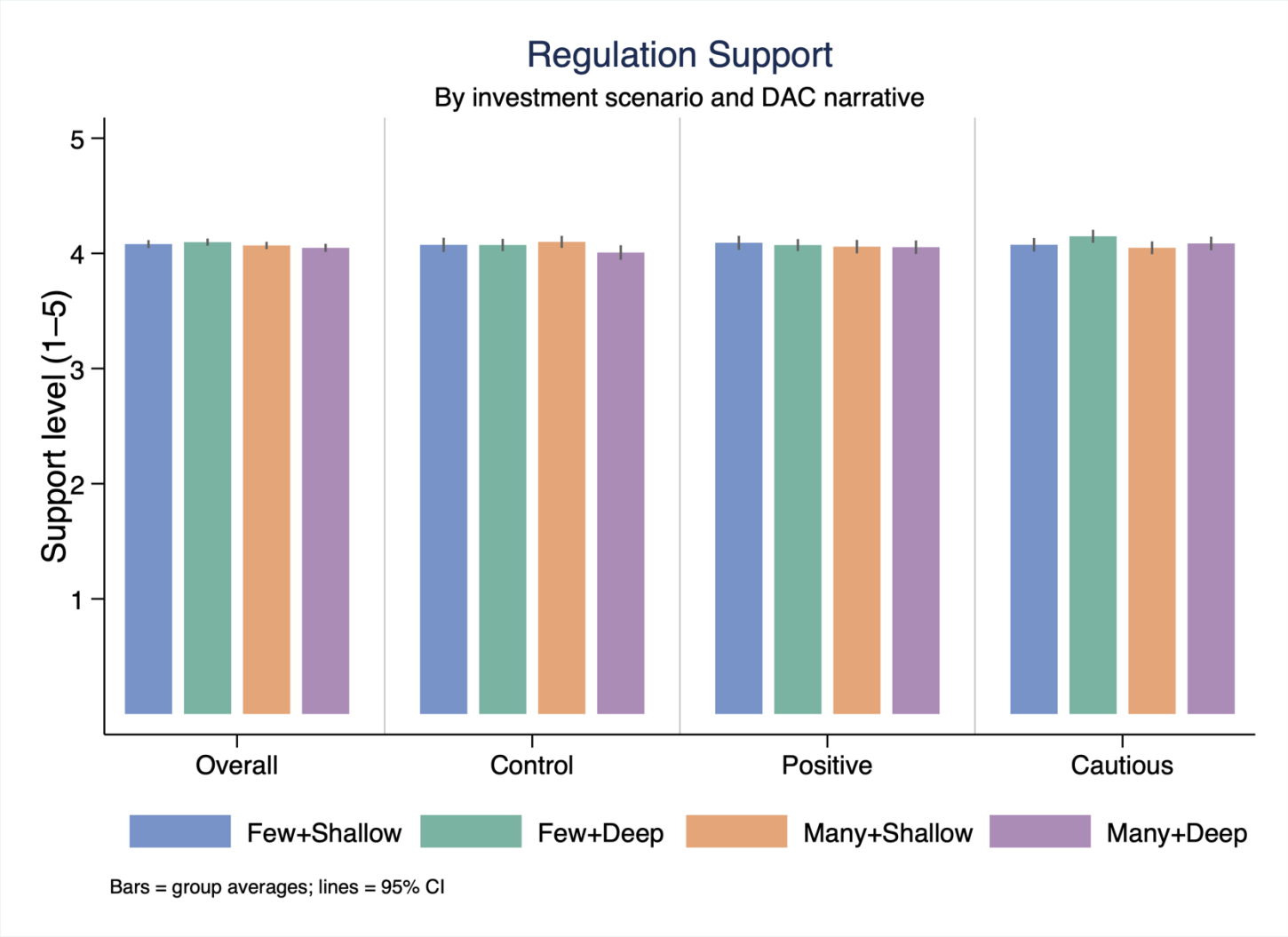
A majority (≈75%) of oil and gas companies invest in carbon removal technologies such as Direct Air Capture (DAC).

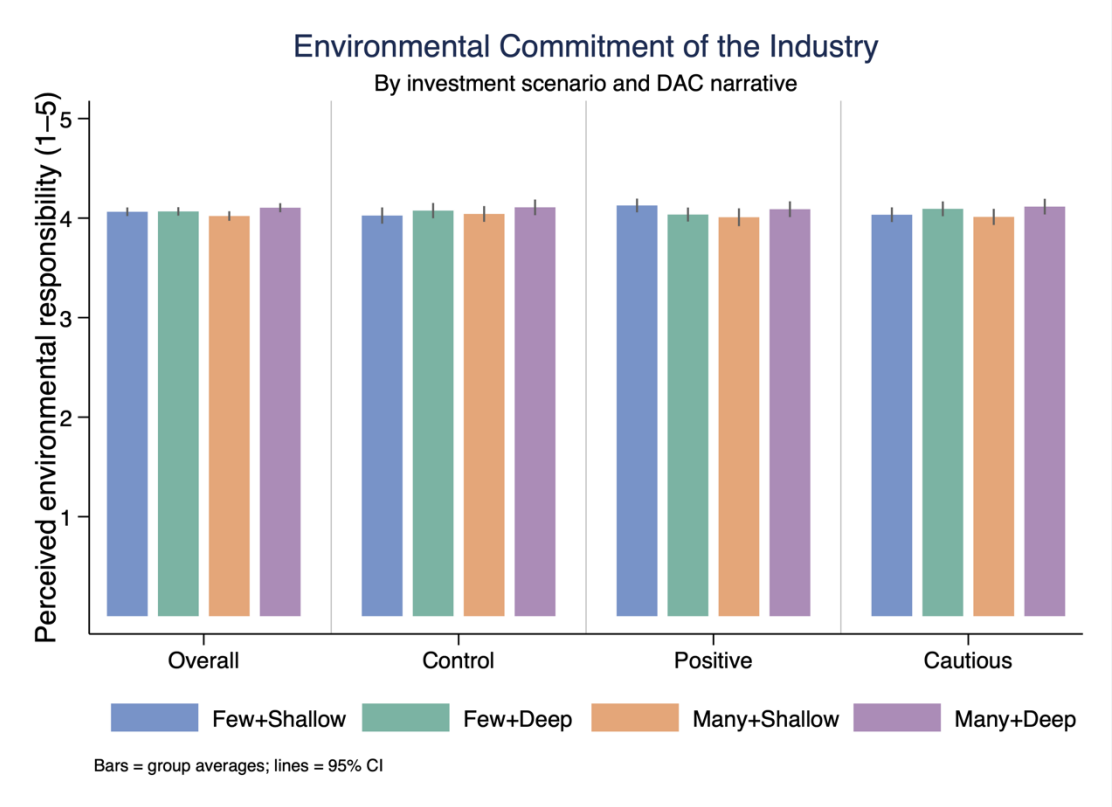
Shallow vs. Deep

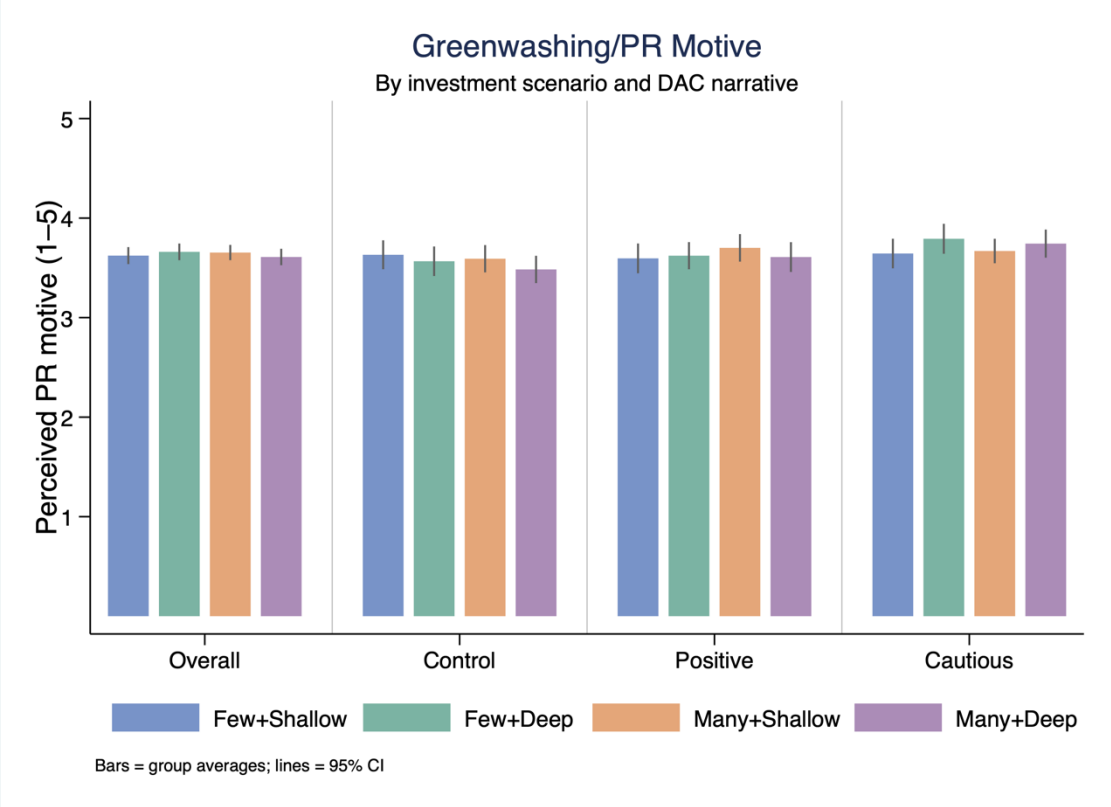
These companies' investments are relatively small, mainly aimed at feasibility assessments and early-stage demonstration projects.

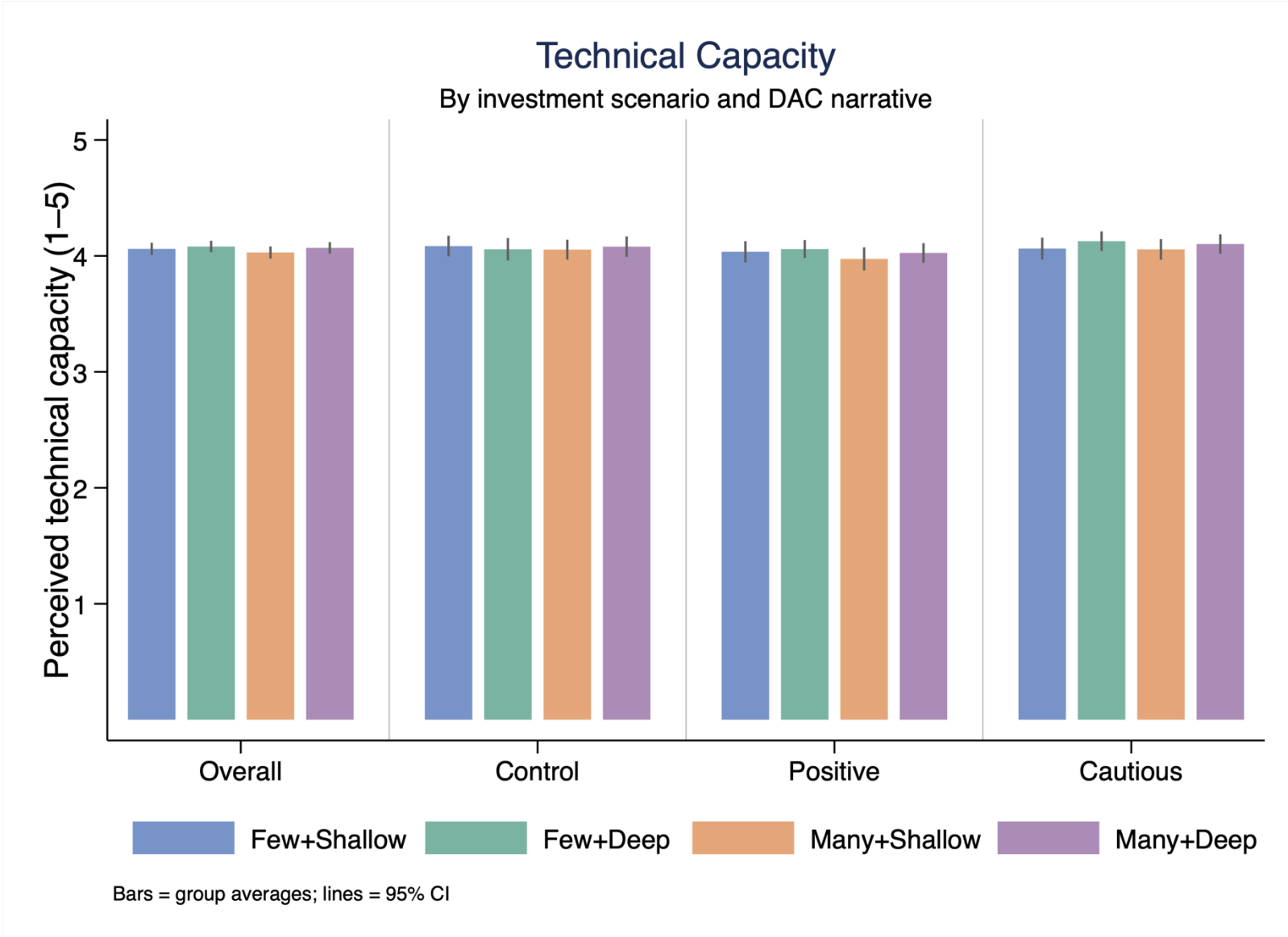
These companies commit to large-scale investments and position carbon removal as a core component of their long-term climate strategies.

- Impose taxes on fossil fuel production or emissions
- Ban new fossil fuel extraction projects
- Gradually phase out fossil fuels over the coming decades
- Eliminate government subsidies for the fossil fuel industry
- Require the fossil fuel industry to regularly disclose greenhouse gas emissions data
- Require the fossil fuel industry to publicly disclose investments in decarbonization technologies and their progress




Environmental Commitment of the Industry
By investment scenario and DAC narrative
Perceived environmental responsibility (1–5)
5
4
3
2
1
Overall
Control
Positive
Cautious
Few+Shallow
Few+Deep
Many+Shallow
Many+Deep
Bars = group averages; lines = 95% CI


Greenwashing/PR Motive
By investment scenario and DAC narrative
Perceived PR motive (1–5)
5
4
3
2
1
Overall
Control
Positive
Cautious
Few+Shallow
Few+Deep
Many+Shallow
Many+Deep
Bars = group averages; lines = 95% CI


Technical Capacity
By investment scenario and DAC narrative
Perceived technical capacity (1–5)
5
4
3
2
1
Overall
Control
Positive
Cautious
Few+Shallow
Few+Deep
Many+Shallow
Many+Deep
Bars = group averages; lines = 95% CI

Conclusions

- **Modest mitigation deterrence:**
  - DAC narratives slightly reduce perceived ambition of conventional mitigation.
- **No evidence of greenwashing:**
  - Corporate DAC investment claims do *not* reduce public support for strict regulation of fossil fuel companies.
- **The larger concern:**
  - Across all narratives, respondents' expectations for emissions reductions fall far short of China's official targets and IPCC-recommended pathways—suggesting a systematic underestimation of mitigation urgency that is *independent of DAC*.